

PROJECT DESCRIPTION / SYSTEM DEVELOPMENT BRIEF

Project Title

Artificial Intelligence Powered Educational Service Assistant for Nigerian Tertiary Institutions

Case Study

The Federal Polytechnic, Ado-Ekiti, Ekiti State

1. PROJECT OVERVIEW

This project involves the design and development of an AI-powered educational service assistant for Nigerian tertiary institutions, using The Federal Polytechnic, Ado-Ekiti as the case study.

The main purpose of the system is to provide students, prospective students, staff, parents, and other stakeholders with a centralized platform where they can obtain accurate information about institutional and administrative services through a conversational Artificial Intelligence assistant.

The system will function as an intelligent chatbot or virtual assistant capable of understanding natural-language questions and providing relevant responses based on a trusted institutional knowledge base.

The project focuses mainly on non-academic administrative support. It is not intended to replace lecturers, classroom teaching, academic assessment, or institutional decision-makers.

2. THE PROBLEM THE SYSTEM IS SOLVING

Students in Nigerian tertiary institutions often experience difficulty obtaining accurate and timely information about institutional services.

Students may have to

- Visit multiple offices physically.
- Search through different websites and portals.
- Ask individual lecturers or administrative staff.
- Depend on unofficial social media information.
- Wait until administrative offices are open.
- Repeatedly ask staff the same questions.

Common areas where students experience difficulties include

- Admission procedures.
- Course registration.
- School fees.
- Examination schedules.
- Academic calendars.
- Hostel allocation.

- SIWES information.
- Library services.
- Transcript processing.
- ICT support.
- Student records.
- Graduation requirements.

The proposed AI-powered assistant will provide a centralized digital platform where users can ask questions and receive relevant responses at any time.

3. MAIN AIM OF THE PROJECT

The main aim is to develop an AI-Powered Educational Service Assistant that improves access to educational and administrative support services in Nigerian tertiary institutions.

The system will use Artificial Intelligence technologies to understand user enquiries and provide relevant information based on verified institutional data.

4. MAIN OBJECTIVES

The system should

- Provide a centralized platform for educational and administrative enquiries.
- Allow users to communicate with the system using natural language.
- Provide information about institutional services.
- Reduce repetitive enquiries handled manually by administrative staff.
- Make institutional information more accessible.
- Provide support outside normal office hours.
- Use a trusted knowledge base to improve response accuracy.
- Demonstrate the application of AI technologies in Nigerian tertiary institutions.

5. TARGET USERS

Students

Students can ask questions relating to course registration, school fees, examination information, academic calendars, SIWES, hostel services, library services, ICT support, transcript information, and graduation requirements.

Prospective Students

Prospective students can ask questions relating to admission requirements, available courses, departments, ND programmes, HND programmes, application procedures, JAMB requirements, and screening information.

Staff

Staff may access institutional information and relevant administrative guidance.

Parents and Guardians

Parents may make general enquiries concerning admissions and institutional services.

System Administrators

Administrators will manage the knowledge base and maintain the system.

6. CORE SYSTEM FEATURES

A. AI CHAT ASSISTANT

This is the major feature of the system.

Users should be able to type questions naturally, for example

- “How do I register my courses?”
- “What are the requirements for admission?”
- “When is the examination timetable available?”
- “How do I apply for hostel accommodation?”
- “What documents do I need for SIWES?”
- “How do I request my transcript?”

The system should process the user's question and provide an appropriate response.

The chat interface should be simple, modern, and easy to use.

B. SERVICE CATEGORIES

The website should organize major institutional services into categories.

1. Admission

Admission requirements, ND programmes, HND programmes, available courses, departments, application procedures, required documents, JAMB information, and screening guidelines.

2. Course Registration

Course registration procedures, registration requirements, registration deadlines, portal guidance, and course selection.

3. School Fees

School fee procedures, payment guidance, payment deadlines, and payment-related requirements.

4. Examinations

Examination schedules, examination requirements, timetables, and academic regulations.

5. Academic Calendar

Resumption, registration periods, examination periods, holidays, and important academic dates.

6. Hostel Services

Hostel applications, hostel allocation, and accommodation requirements.

7. SIWES

SIWES requirements, registration, placement, documentation, and procedures.

8. Library Services

Library registration, borrowing materials, library rules, and digital resources.

9. ICT Support

Student portal support, login problems, technical assistance, and institutional digital services.

10. Transcript Services

Transcript requests, required procedures, and required documents.

11. Graduation Requirements

Graduation procedures, clearance, required documents, and institutional requirements.

7. KNOWLEDGE BASE

The AI assistant should not simply provide random answers.

It should use a verified institutional knowledge base containing trusted information.

The knowledge base may include

- Frequently asked questions.
- Institutional policies.
- Administrative procedures.
- Service information.
- Academic calendars.
- Admission information.
- Departmental information.
- Official documents.

The administrator should be able to manage and update this information.

Possible knowledge base fields include

- Question.
- Answer.
- Category.
- Keywords.
- Source.
- Last updated date.
- Status.

8. AI TECHNOLOGIES INVOLVED

Natural Language Processing (NLP)

NLP enables the system to understand and process human language. The user should not need to use specific commands.

For example, the following questions should ideally be understood as having a similar meaning

- “How do I register courses?”
- “Course registration process?”
- “How can I register my courses?”
- “What is the procedure for registering courses?”

Large Language Models (LLMs)

An LLM may be used to

- Understand user questions.
- Generate conversational responses.
- Provide explanations.
- Maintain context.

However, the LLM should preferably use trusted institutional information rather than inventing answers.

Retrieval-Augmented Generation (RAG)

RAG is an important concept for the project.

Expected process

1. User asks a question.
2. The system searches the institutional knowledge base.
3. Relevant information is retrieved.
4. The AI uses the retrieved information to generate a response.
5. The answer is presented to the user.

This approach is intended to improve accuracy and ensure that responses are based on approved information.

9. PROPOSED WEBSITE STRUCTURE

1. Home Page

The homepage should introduce the platform.

Possible sections

- Navigation bar.
- Hero section.
- Introduction to the AI assistant.
- “Ask the Assistant” call-to-action.
- Service categories.
- How the system works.
- Key benefits.

- Frequently asked questions.
- Footer.

2. AI Chat Page

This is the main system page.

Features

- Chat messages.
- User input field.
- Send button.
- Suggested questions.
- Service category shortcuts.
- Conversation history.
- AI typing/loading indicator.

3. Services Page

This page will display the available support categories, including Admissions, Course Registration, School Fees, Examination, Hostel, SIWES, Library, ICT Support, and Transcript services.

4. Knowledge Base / Information Page

This page may display verified institutional information and frequently asked questions.

5. Login/Register Page

This may be included if the system requires user accounts.

Possible features

- Student registration.
- Login.
- User profile.
- Saved conversations.

6. Admin Dashboard

The administrator should be able to

- Add new knowledge base information.
- Edit existing information.
- Delete outdated information.
- Manage service categories.
- View user enquiries.
- View AI conversations.
- Monitor frequently asked questions.

10. SYSTEM WORKFLOW

Basic workflow

User → Website → AI Assistant → Knowledge Base Search → Relevant Information → AI Response → User

Detailed process

1. The user opens the website.
2. The user enters a question.
3. The system receives the question.
4. The AI processes and understands the question.
5. The system identifies the relevant service or topic.
6. The system searches the institutional knowledge base.
7. Relevant information is retrieved.
8. The AI generates a response based on the information.
9. The response is displayed to the user.

11. POSSIBLE TECHNOLOGY STACK

Frontend

- React.js
- Vite
- Tailwind CSS or regular CSS

Backend

- Node.js
- Express.js

Database

- MongoDB

AI Integration

- LLM API
- OpenAI API or another suitable AI API

RAG / Knowledge Base

Possible components may include

- Document storage.
- Embeddings.
- Vector search.
- MongoDB Vector Search or another vector database.

12. DATABASE STRUCTURE

Users

- Name
- Email
- Password
- Role
- Created date

Categories

- Category name
- Description
- Icon

Knowledge Base

- Title
- Question
- Content
- Category
- Keywords
- Source
- Date updated
- Status

Conversations

- User ID
- Messages
- Date
- Conversation title

Enquiries / Feedback

- User question
- AI response
- Feedback rating
- Status
- Date

13. ADMINISTRATOR RESPONSIBILITIES

The administrator should be able to

- Log into the admin dashboard.
- Add new institutional information.
- Update existing information.
- Delete outdated information.
- Manage categories.
- Review common user questions.
- Monitor conversations.
- Improve the knowledge base.

This is important because institutional information can change. The AI system should therefore be supported by regularly updated information.

14. DESIGN REQUIREMENTS

The website should look

- Modern.
- Professional.
- Clean.
- Educational.
- AI-focused.
- Responsive.
- Mobile-friendly.
- Easy to navigate.

The interface should feel like a combination of an institutional website, a modern AI application, and a student support platform.

The AI assistant should be the main focus of the user experience.

15. MAIN BENEFITS OF THE SYSTEM

The proposed system is expected to provide

24/7 Availability

Users can access information outside normal office hours.

Faster Responses

Students can receive immediate responses to common questions.

Centralized Information

Institutional information can be provided from a single platform.

Reduced Administrative Workload

The system can handle repetitive enquiries automatically.

Improved Accessibility

Students do not always need to visit multiple offices physically.

Improved Student Experience

The platform provides a simpler way to access institutional information.

16. IMPORTANT LIMITATIONS AND RULES

The system should not

- Replace lecturers.
- Replace classroom teaching.
- Make sensitive academic decisions without human oversight.
- Invent institutional information.
- Provide unverified information as official information.

When the AI does not have sufficient information, it should indicate that clearly.

Example

"I could not find verified information for this question. Please contact the appropriate department or check the official institutional channel."

17. MINIMUM VIABLE PRODUCT (MVP)

For the first version of the project, the system should include

1. A professional homepage.
2. An AI chat interface.
3. Service categories.
4. A knowledge base containing institutional information.
5. AI-powered responses.
6. An admin dashboard.
7. Add/edit/delete knowledge base information.
8. Responsive design.

18. FUTURE FEATURES

Possible future improvements include

- Voice input.
- Voice responses.
- Multilingual support.
- Yoruba language support.
- WhatsApp integration.

- SMS integration.
- Mobile application.
- Student portal integration.
- Personalized responses.
- User authentication.
- Conversation history.
- Human support escalation.
- Analytics dashboard.
- Feedback and rating system.

FINAL PROJECT UNDERSTANDING

This project is a web-based AI-powered educational service assistant designed for Nigerian tertiary institutions, with The Federal Polytechnic, Ado-Ekiti as the case study.

The core of the system is an intelligent conversational assistant that allows users to ask questions about institutional and administrative services using normal language.

The system uses AI technologies such as Natural Language Processing, Large Language Models, and potentially Retrieval-Augmented Generation to provide responses based on a verified institutional knowledge base.

The main goal is to provide a centralized, intelligent, accessible, and efficient platform for answering educational and administrative enquiries while reducing the need for students to physically visit multiple offices or depend on unofficial information sources.

The system should be built as a professional, responsive web application with a modern frontend, backend, database, AI integration, and administrative knowledge-base management system.